

Pre-Existing Conditions and Adverse Outcomes of COVID-19

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Motivation & Goal

MOTIVATION



Personal Connection

Loss of a family member with diabetes & cardiovascular disease while living with my senior mother.



Academic Interest

Sparked passion for data science and public health analysis to solve real-world problems.



Research Relevance

COVID-19 highlights how pre-existing conditions severely impact health outcomes.



Future Risk Mitigation

Ongoing risk of future pandemics reinforces the need to focus on underlying conditions.

THE GOAL

Identify risk factors to support individual awareness, policy decisions, and future research.

Research Questions

1. Risk Factors by Age Group

Which demographics, health behaviors, and pre-existing conditions are most associated with severe COVID-19 outcomes across age groups?

2. Model Performance Comparison

How do classification models (Logistic Regression & Random Forest) compare in predicting severe outcomes?

3. Comorbidity Interactions

Do combinations of pre-existing conditions increase the risk of severe outcomes, and how does this vary by age group?

Methodology

CLASSIFICATION MODELS

Logistic Regression

Four Models Developed:

- Full dataset vs. Reduced (near-zero variance removed)
- With and without interaction effects

Random Forest

Four Models Developed:

- Full dataset vs. Reduced (near-zero variance removed)
- Untuned and after tuning

Classification models were utilized to compare performance in predicting severe COVID-19 outcomes and interpreting risk factors.

Methodology

UNSUPERVISED LEARNING

Association Rule Mining (Market Basket Analysis)

Approach:

- Rules generated with **Adverse Outcome** as the right-hand side (RHS) (consequents)
- Identifies frequent **patterns** and **co-occurrences** within the COVID-19 dataset
- Aims to discover **strong associations** between left-hand side (LHS) (antecedents) and adverse outcomes

This unsupervised approach complements supervised models by highlighting key feature combinations linked to patient risk.

Dataset

Source & Population

Mexico Government COVID-19 dataset
via Translated and Partially uploaded version on Kaggle

Population: Lab-tested COVID-19 cases 2020-2021
via health-center medical units

Dimensions

Size:

~1M obs → ~388K (cleaned)

Variables:

21 original → 14 (Full cleaned)
→ 8 (Near-zero removed)

Target Variable: Adverse Outcome

Defined as ICU admission, intubation, or death.

Dataset Features

Target Variable: Adverse Outcome

Defined as ICU admission, intubation, or death.

Demographics



Age and Age Group

(<20, 20-35, 36-50, 51-64, 65-80, 81+)



Sex

Health Behaviors



Tobacco Use



Pregnancy

Dataset Features

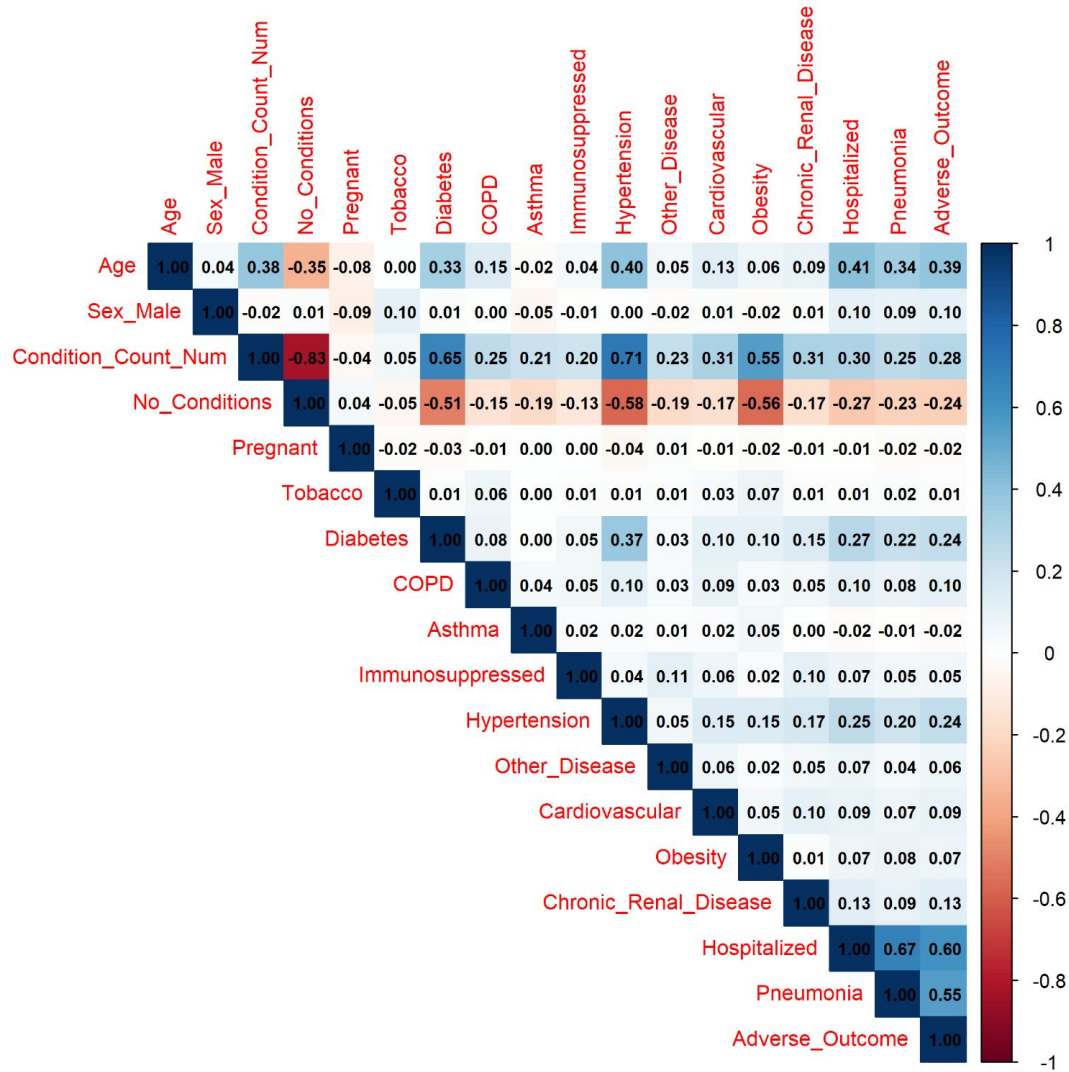
Pre-existing Conditions

- ✓ Diabetes
- ✓ Hypertension (High blood pressure)
- ✓ Obesity
- ✓ COPD (Chronic obstructive pulmonary disease)
- ✓ Asthma
- ✓ Immunosuppression
- ✓ Cardiovascular Disease
- ✓ Chronic Renal Disease (Kidney)
- ✓ "Other Conditions"

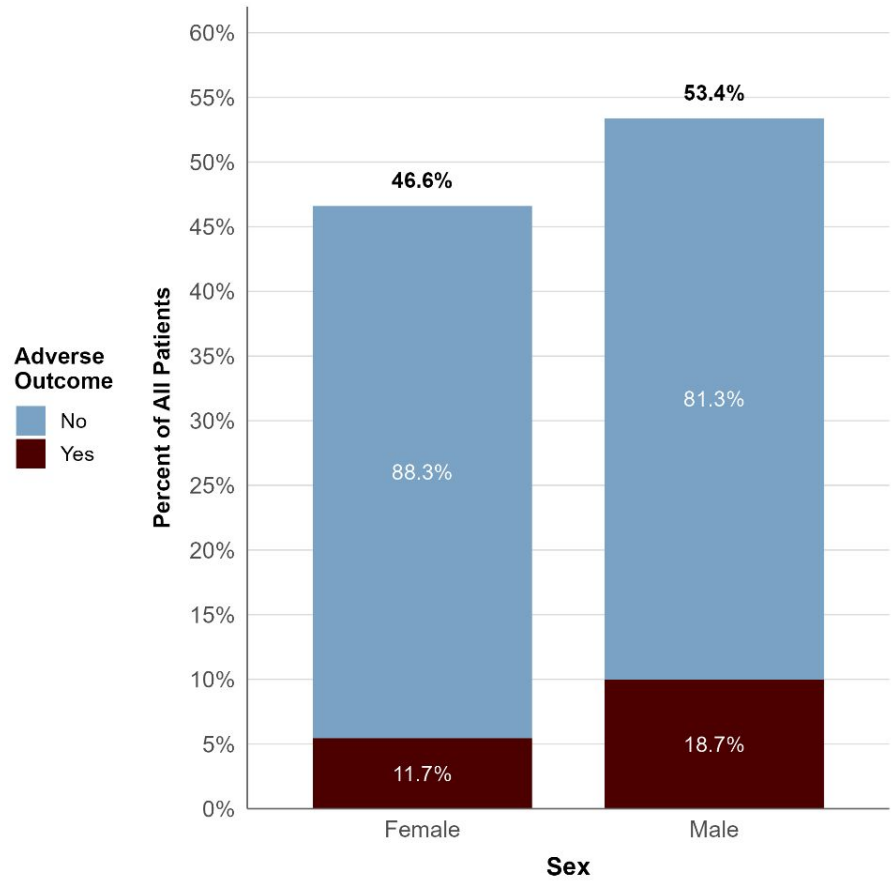
Aggregate Feature: Conditions Count (0-4+)

EDA & Initial Results

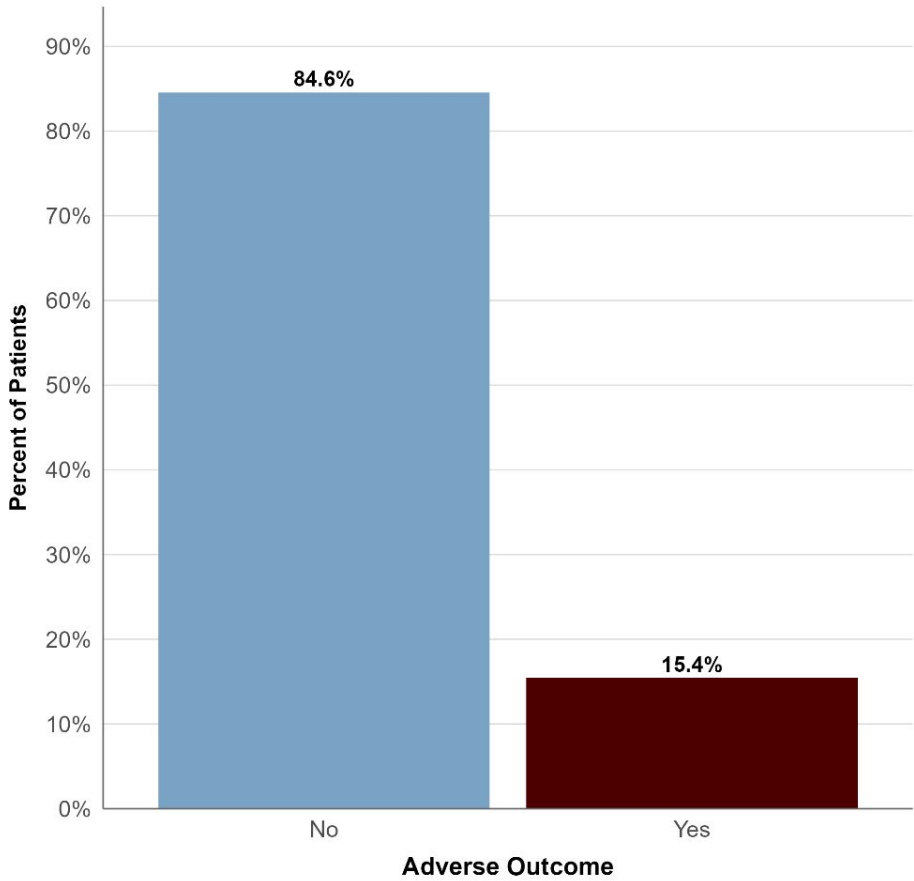
Correlation



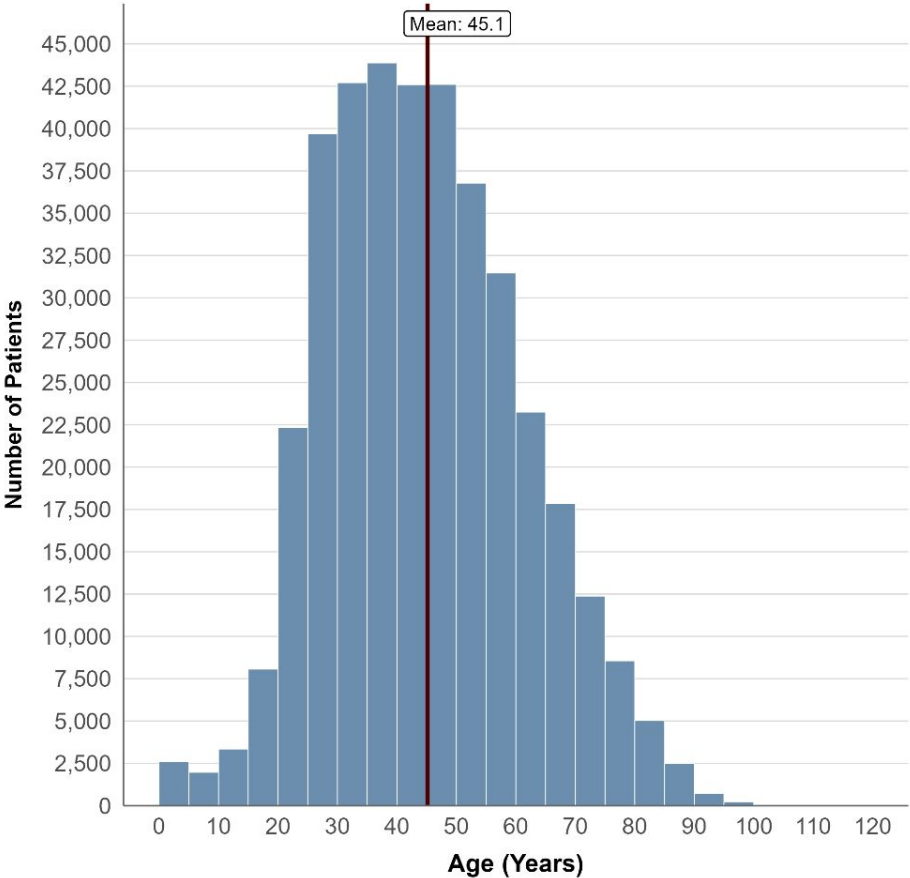
Adverse Outcome by Sex



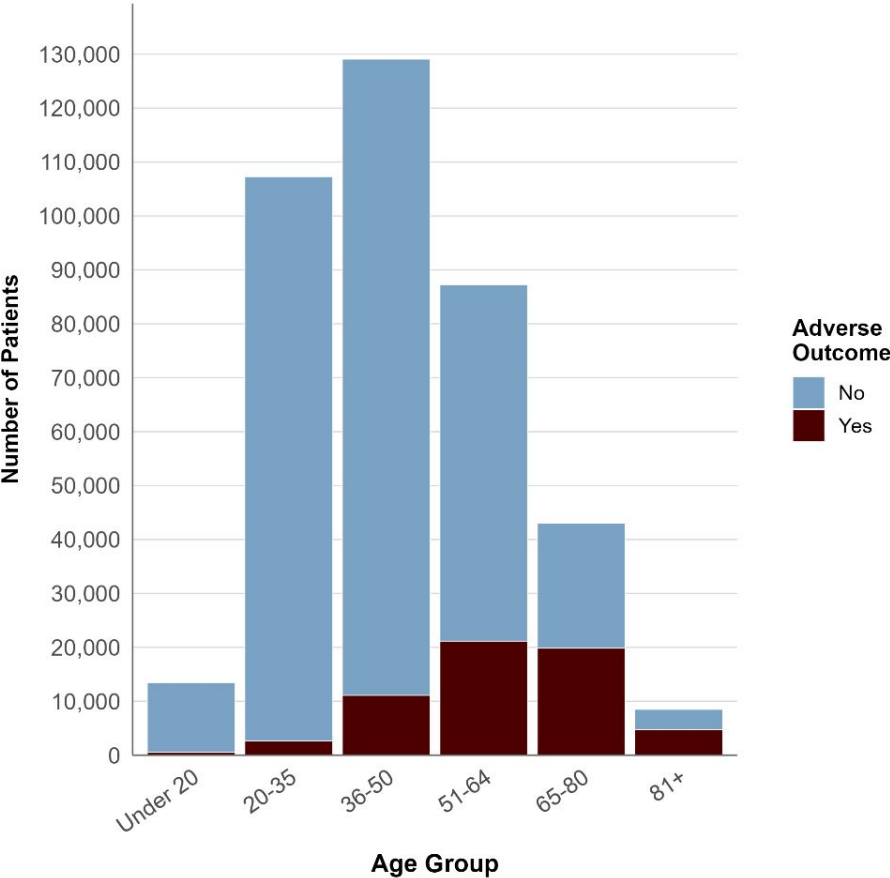
Overall Adverse Outcome



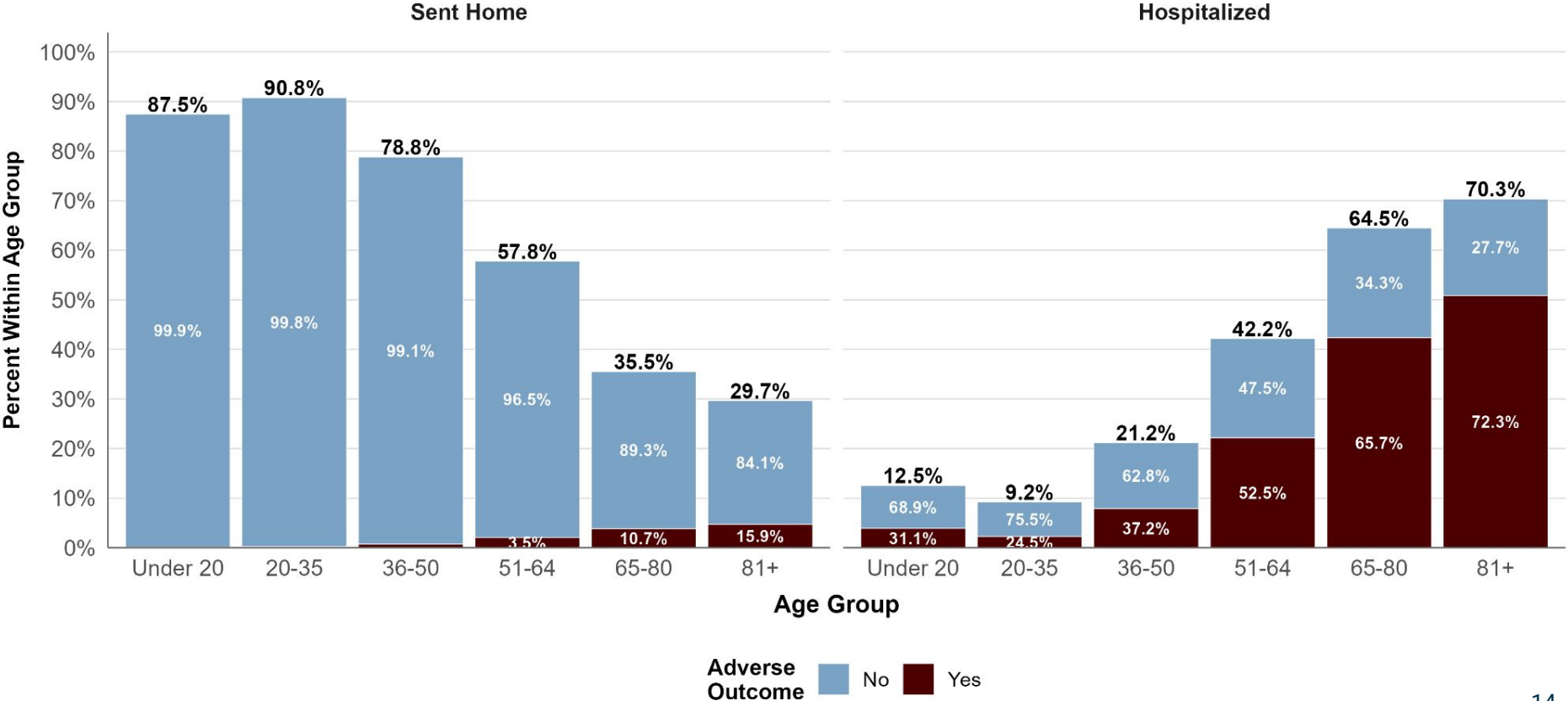
Age Distribution



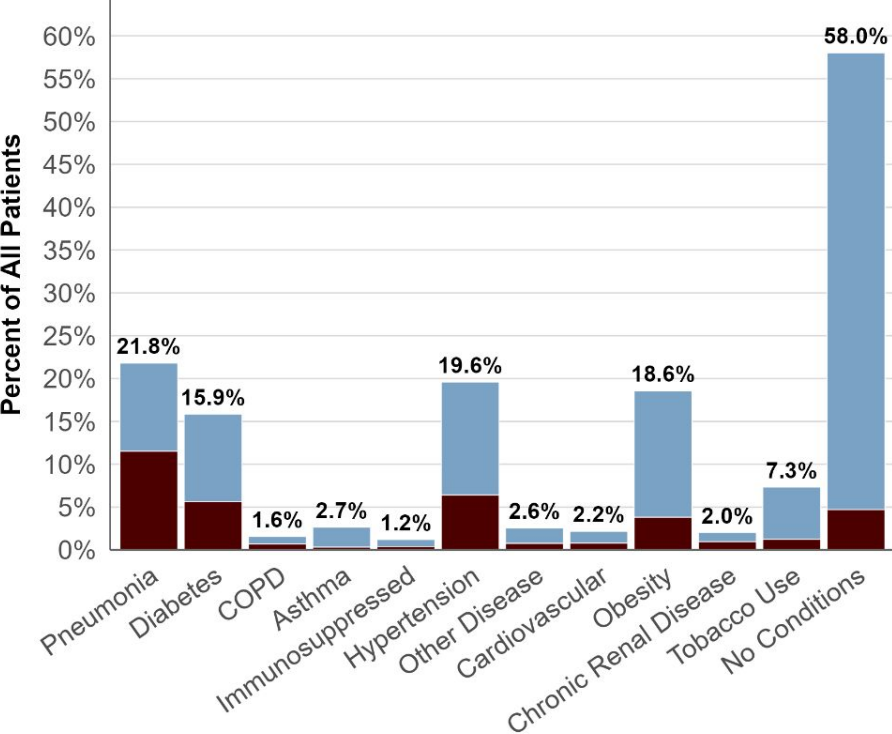
Adverse Outcomes by Age Group



Hospitalization by Age Group

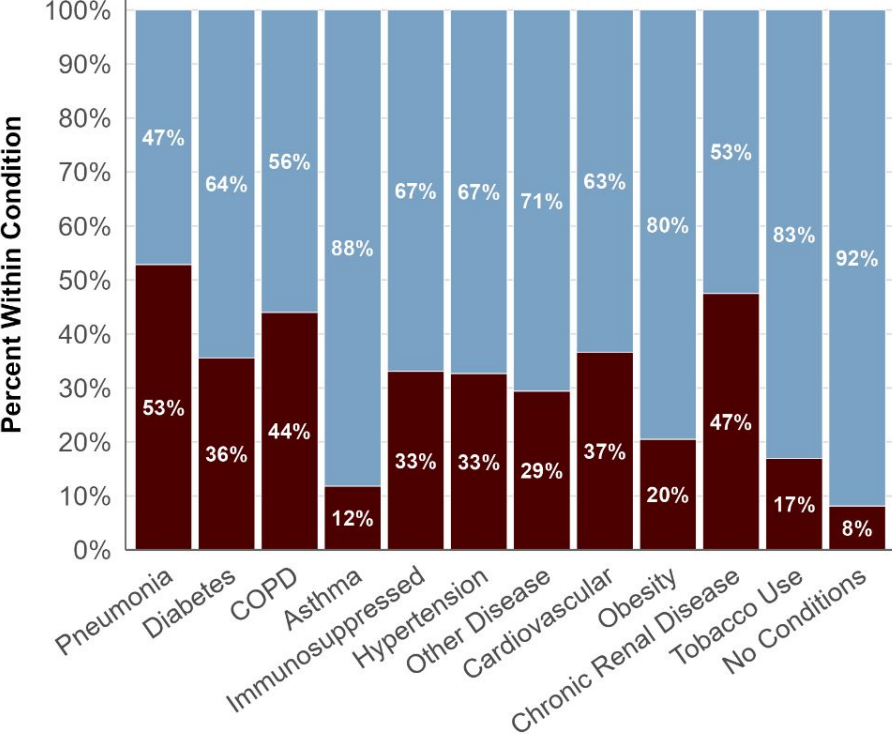


Condition Prevalence



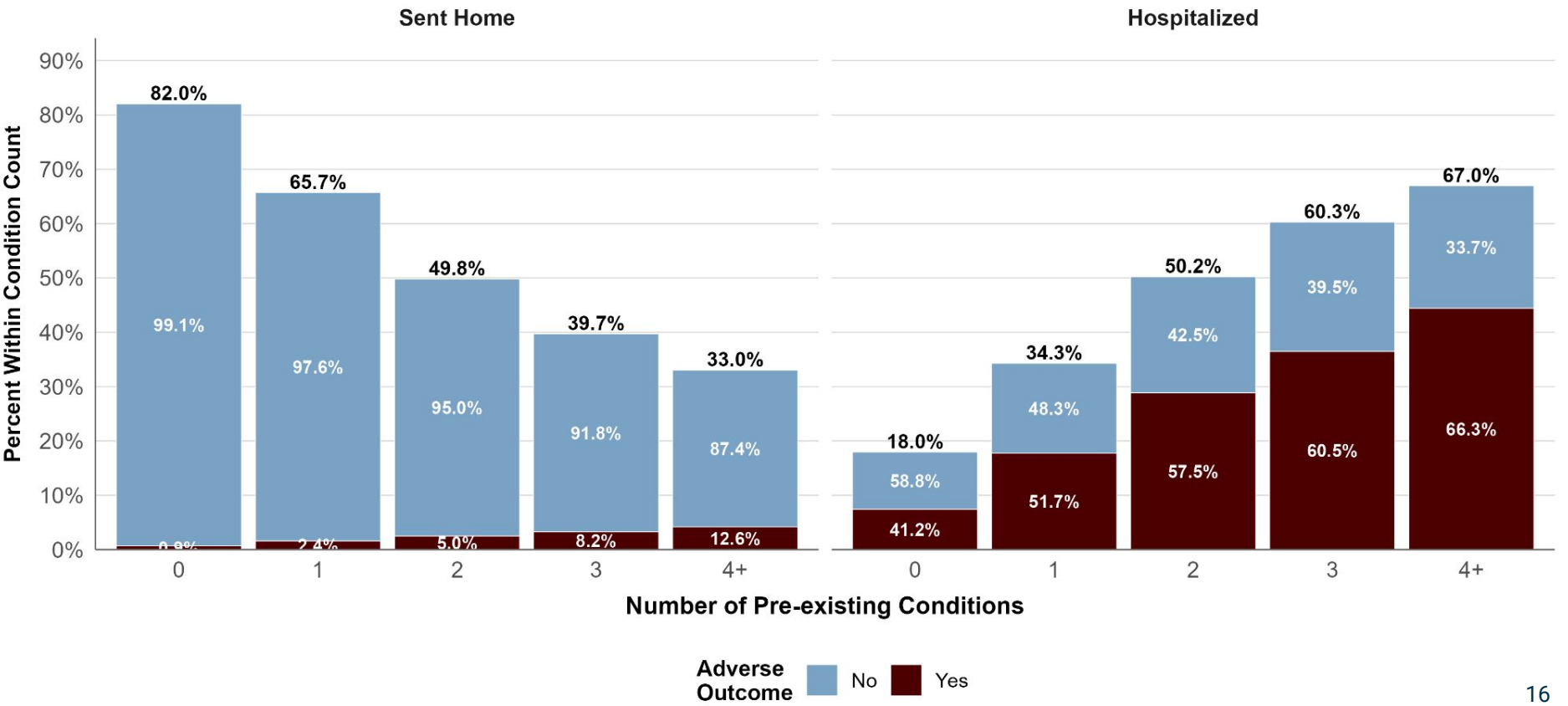
Adverse Outcome No Yes

Adverse Outcome Rate



Adverse Outcome No Yes

Hospitalization by Condition Count



Association Rule Mining

Unsupervised Model

Association Rule Mining

Overview & Rule Structure

Used for **unsupervised** pattern discovery

Identifies **co-occurring conditions** linked to outcomes

Not designed for prediction

Also known as **market basket analysis**

Rule Structure

LHS (X)

Antecedents

Condition(s)



RHS (Y)

Consequents

Outcome (Adverse)

Key Metrics

Association Rule Mining

Support

$$P(X \cap Y)$$

% of data where both condition(s) and outcome occur

Measures how **common** the rule is

Confidence

$$P(Y | X)$$

Probability of outcome given condition(s)

Measures **risk** within the subgroup

Lift

$$P(Y | X) / P(Y)$$

Compares rule to baseline rate (~15%)

Lift > 1: Higher than expected risk

Lift ≈ 1: No association

$$\frac{P(X \cap Y)}{P(X) \cdot P(Y)}$$

Example: Rule Analysis

Association Rule Mining

{81+ Diabetic Male} →
{Adverse Outcome}

SUPPORT

~0.2% (rare but meaningful)

CONFIDENCE

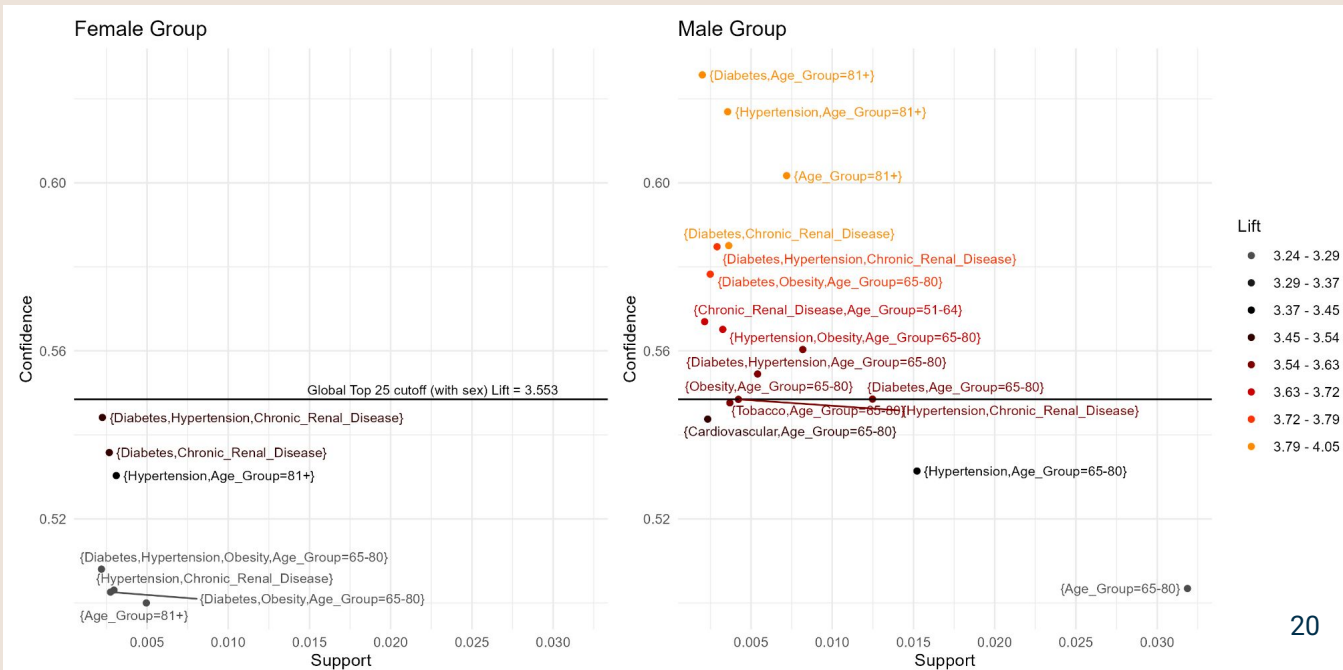
~62.6% (high risk group)

LIFT

~4.1 (4.1× baseline)

ARM thresholds: Support $\geq 0.2\%$, Confidence $\geq 50\%$, Lift ≥ 3.0

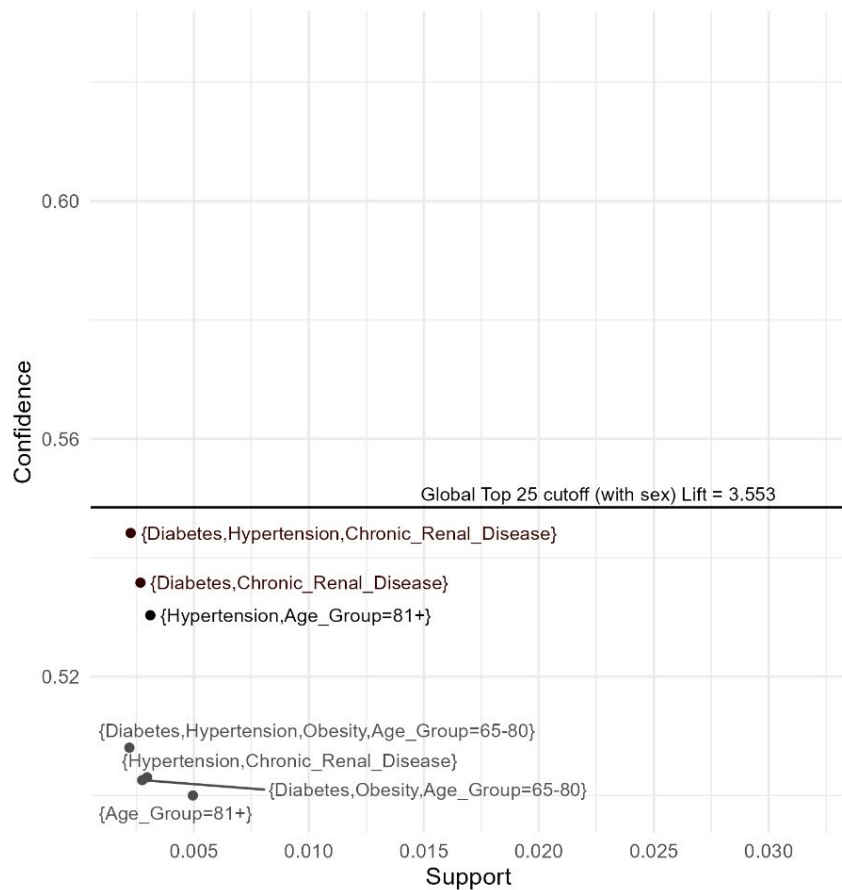
Rules: 52 with sex included, 29 with sex excluded



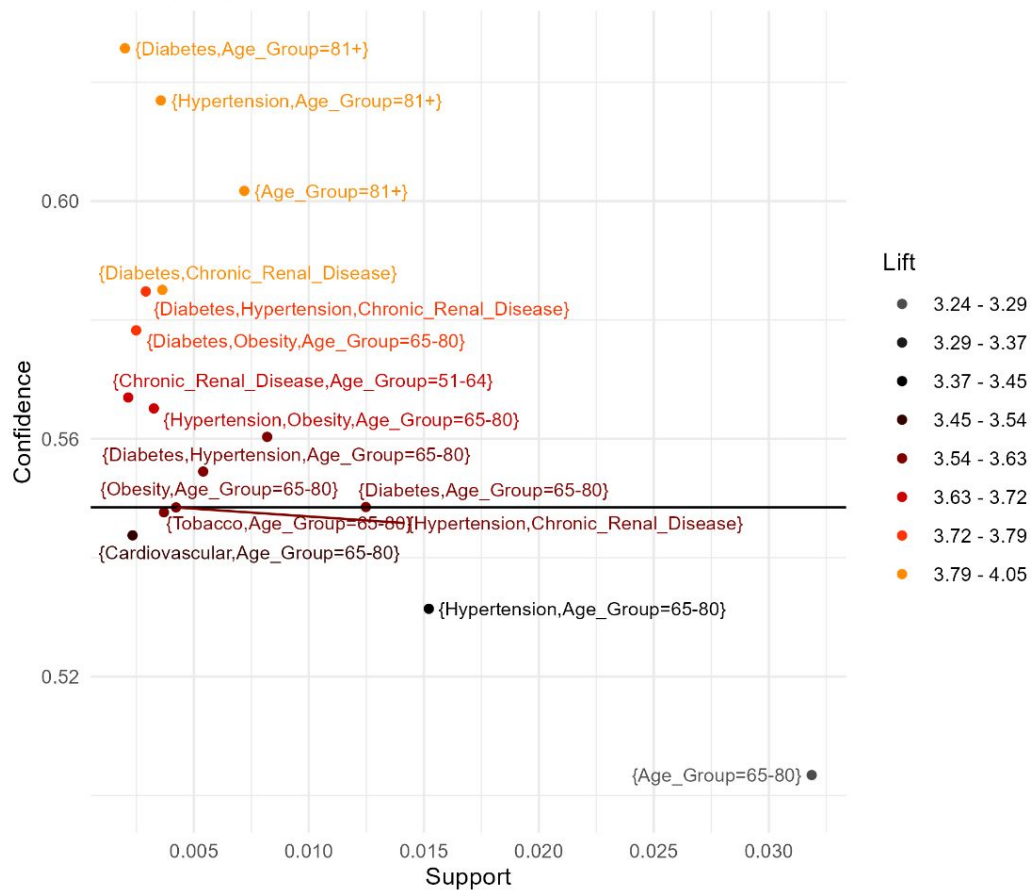
Results & Insights

Association Rule Mining

Female Group



Male Group



Association Rules: Sex Rules

♂ Male Patterns

- Appeared in more high-lift rules than women.
- Lift values generally range from **3 to 4**.
- Tobacco appeared specifically in the **65–80 age group**.
- Included most female patterns but with higher lift metrics.

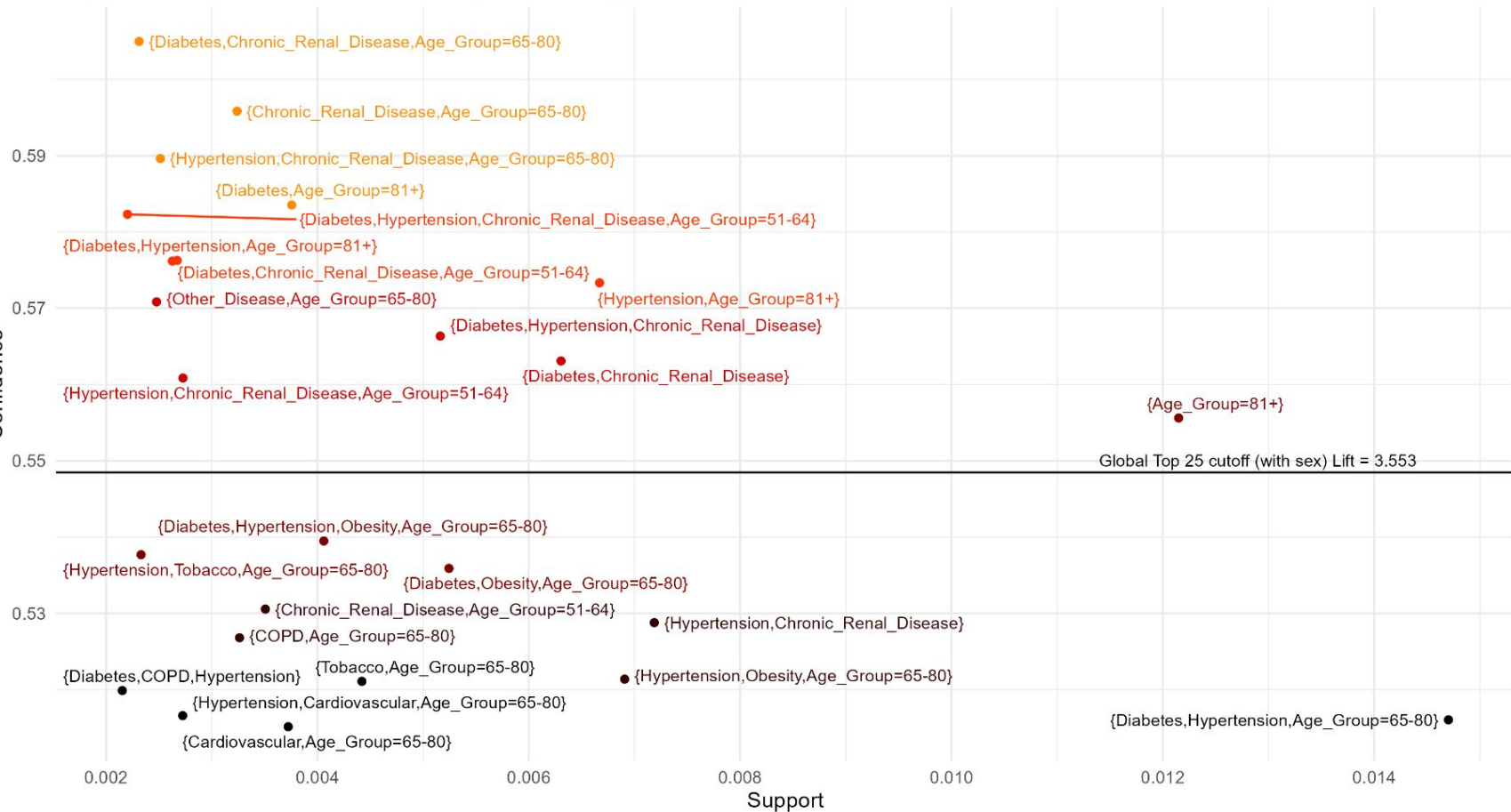
♀ Female Patterns

- Highest recorded rule lift was approximately **3.55**.
- Patterns largely mirrored male rules but exhibited **lower lift overall**.
- Diabetes and hypertension were the most common components.

Key takeaway: Male sex, older age, and comorbidities together showed the strongest rule patterns.

Top 25 Rules for Adverse Outcome (without sex)

Confidence



Lift

- 3.34 - 3.38
- 3.38 - 3.44
- 3.44 - 3.6
- 3.6 - 3.7
- 3.7 - 3.77
- 3.77 - 3.92

Sex-Excluded Rule Patterns

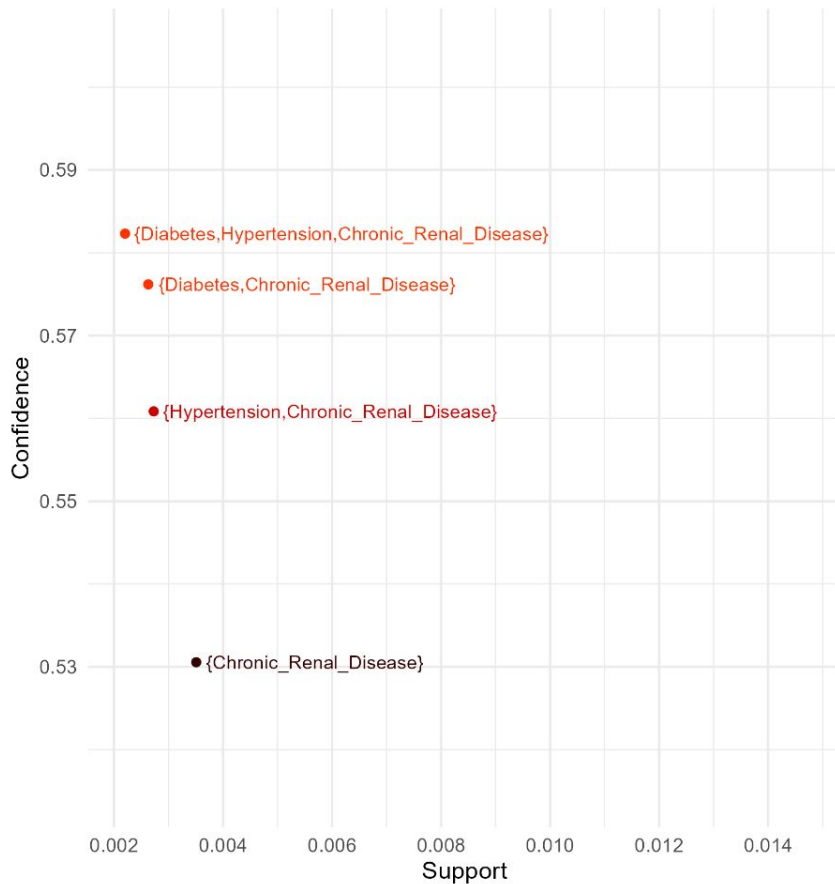


Key Patterns

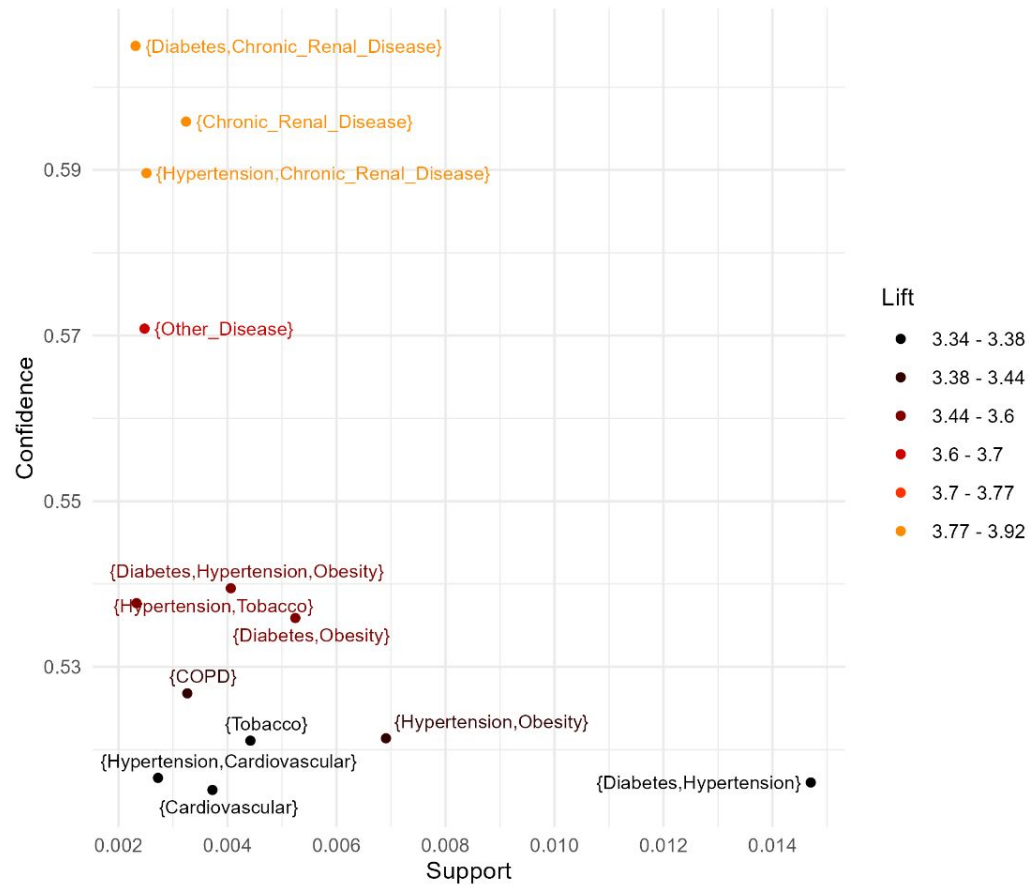
- **Chronic renal disease** dominated the sex-excluded rules.
- **Ages 65-80**: Strongest rules appeared here, with lift around **3.86-3.92**.
- **Ages 81+**: Slightly lower lift, around **3.60-3.78**.
- **Ages 51-64**: Often associated with chronic renal disease plus diabetes or hypertension.
- **Tobacco**: Appeared among ages 65-80 after sex was excluded.

Key takeaway: Even without sex, the strongest rule patterns were driven by age group, chronic renal disease, diabetes, and hypertension.

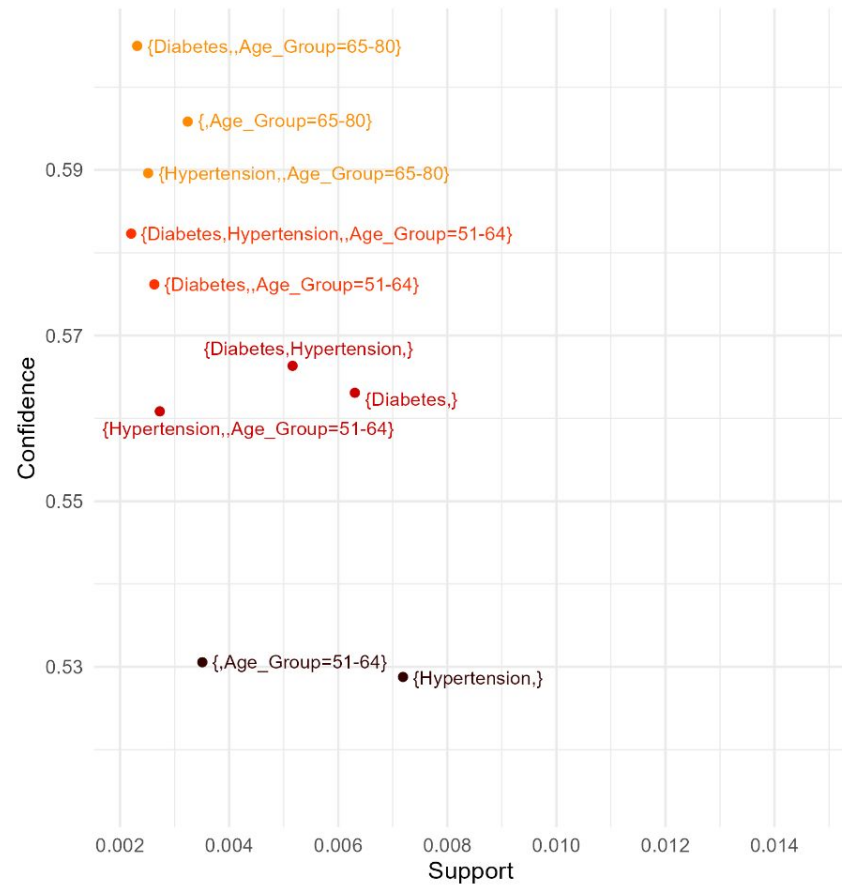
51-64 Age Group



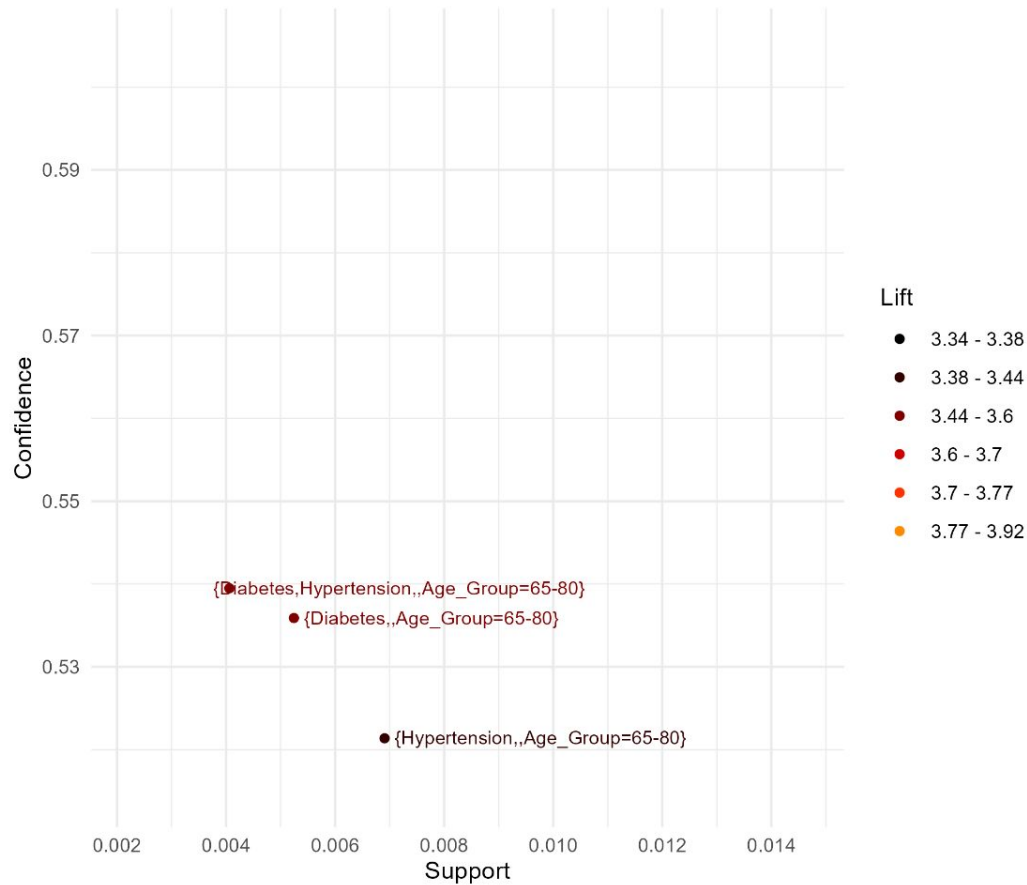
65-80 Age Group



Chronic Renal Disease



Obesity Group



ARM Conclusion

Key takeaway:

Chronic renal disease showed stronger and more varied rule patterns than obesity.



Main Patterns

Chronic Renal Disease: Appeared with diabetes, hypertension, and ages 51-64 or 65-80. Strongest lift near **3.9**.

Obesity: Appeared only in ages 65-80 with diabetes, hypertension, or both. Lift was lower, around **3.4-3.5**.



Modeling Use

ARM guided selected logistic regression interactions

Model Comparison & Evaluation

Supervised Models (LR & RF)

Data Preparation & Modeling Setup

Overview of data splitting, sampling strategies, and variable selection.



80/20 Train/Test Split



Class Imbalance (15%)

Hybrid sampling technique applied to training data:

- Oversample minority $\times 2$
- Undersample majority $\rightarrow$ **final 2:1 ratio**
- Test set remains **completely unchanged**



Output Datasets

Two versions created for model building and evaluation:

Full: 14 variables

Reduced: 8 variables (Near-Zero Variance removed)

Logistic Regression



Model Setup

Built 4 models (Full vs. Reduced, with/without interaction terms).
Selected for **interpretability** and common use in prior research.



Model Specification

Included key interactions:

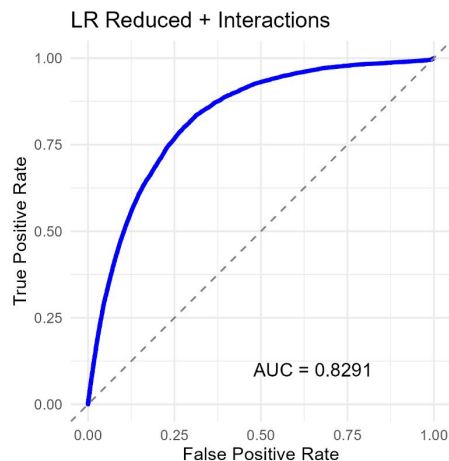
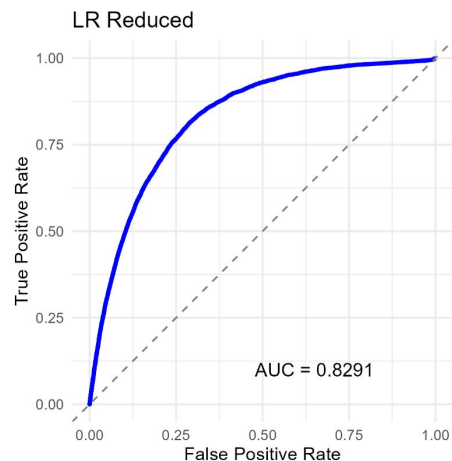
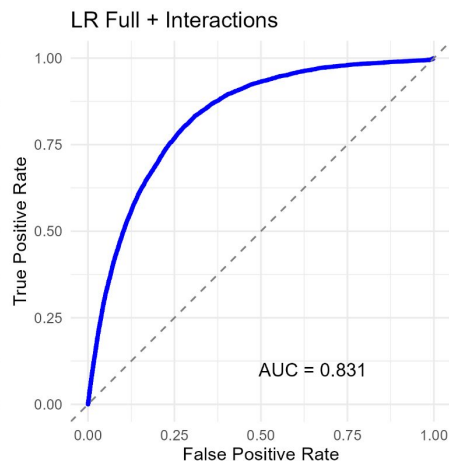
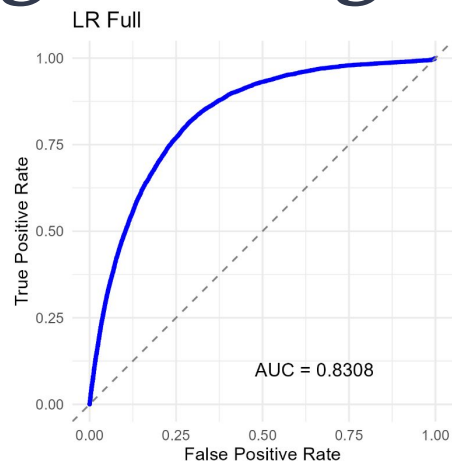
- Diabetes X Hypertension, Sex X Diabetes
- Sex X Hypertension, Sex X Age
- Tobacco X Age, Other_Disease X Age

Excluded:

Diabetes X Obesity (due to high p-value)

Takeaway: Strong interpretability, but limited in capturing complex non-linear relationships.

Logistic Regression



Confusion Matrix and Statistics

Reference

Prediction	No	Yes
No	63693	9444
Yes	2026	2552

Accuracy : 0.8524

95% CI : (0.8499, 0.8549)

No Information Rate : 0.8456

P-value [Acc > NIR] : 7.684e-08

Kappa : 0.2434

Mcnemar's Test P-value : < 2.2e-16

Sensitivity : 0.21274

Specificity : 0.96917

Pos Pred value : 0.55745

Neg Pred value : 0.87087

Precision : 0.55745

Recall : 0.21274

F1 : 0.30795

Prevalence : 0.15436

Detection Rate : 0.03284

Detection Prevalence : 0.05891

Balanced Accuracy : 0.59095

'Positive' class : Yes

Random Forest



Model Setup

Built 2 models: Full vs. reduced dataset

Selected for nonlinear modeling and improved predictive performance



Model Specification

Captures interactions automatically

Provides variable importance for interpretation

Tuning included grid search and using OOB error for best parameters

Key Results

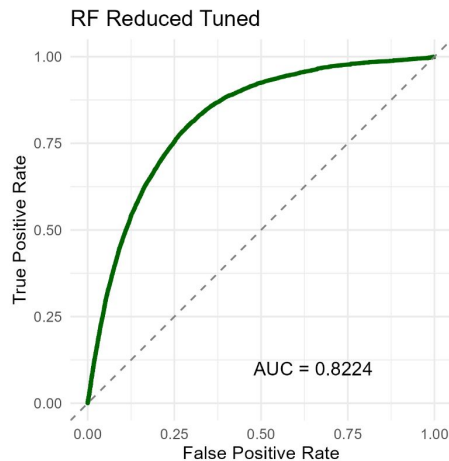
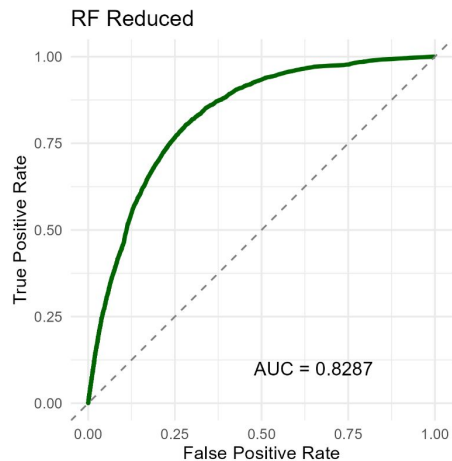
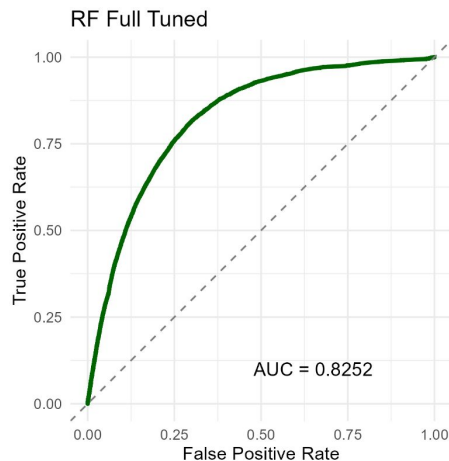
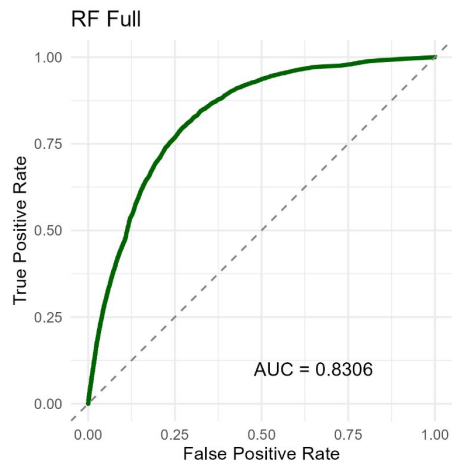
Outperformed logistic regression in **F1** and **sensitivity**

Better at detecting **minority** (adverse outcome) cases

Tuning showed **minimal** improvement

***Takeaway:** Strong predictive performance and interaction capture, but less interpretable than logistic regression*

Random Forest



Confusion Matrix and Statistics

Reference		
Prediction	No	Yes
No	57043	5307
Yes	8676	6689

Accuracy : 0.8201
95% CI : (0.8174, 0.8228)
No Information Rate : 0.8456
P-Value [Acc > NIR] : 1

Kappa : 0.3818

Mcnemar's Test P-value : <2e-16

Sensitivity : 0.55760
Specificity : 0.86798
Pos Pred Value : 0.43534
Neg Pred Value : 0.91488
Precision : 0.43534
Recall : 0.55760
F1 : 0.48894
Prevalence : 0.15436
Detection Rate : 0.08607
Detection Prevalence : 0.19771
Balanced Accuracy : 0.71279

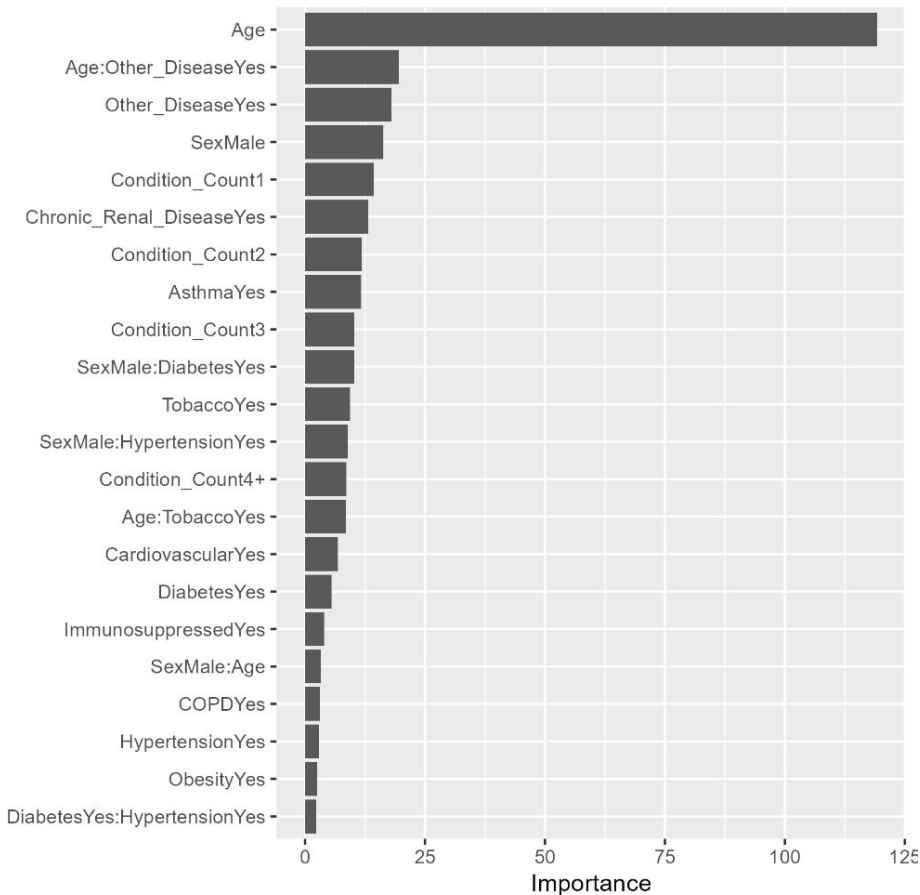
'Positive' Class : Yes

Interpretation of Results

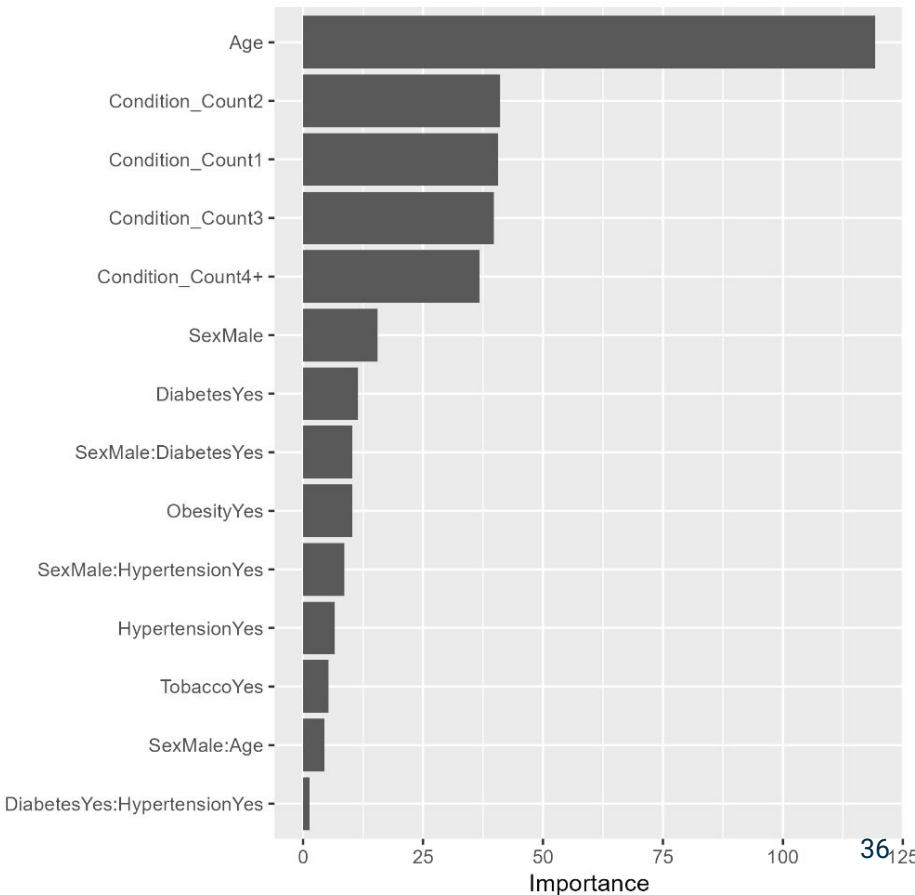
Supervised Models

Logistic Regression

7.3.1 LR - Full Dataset

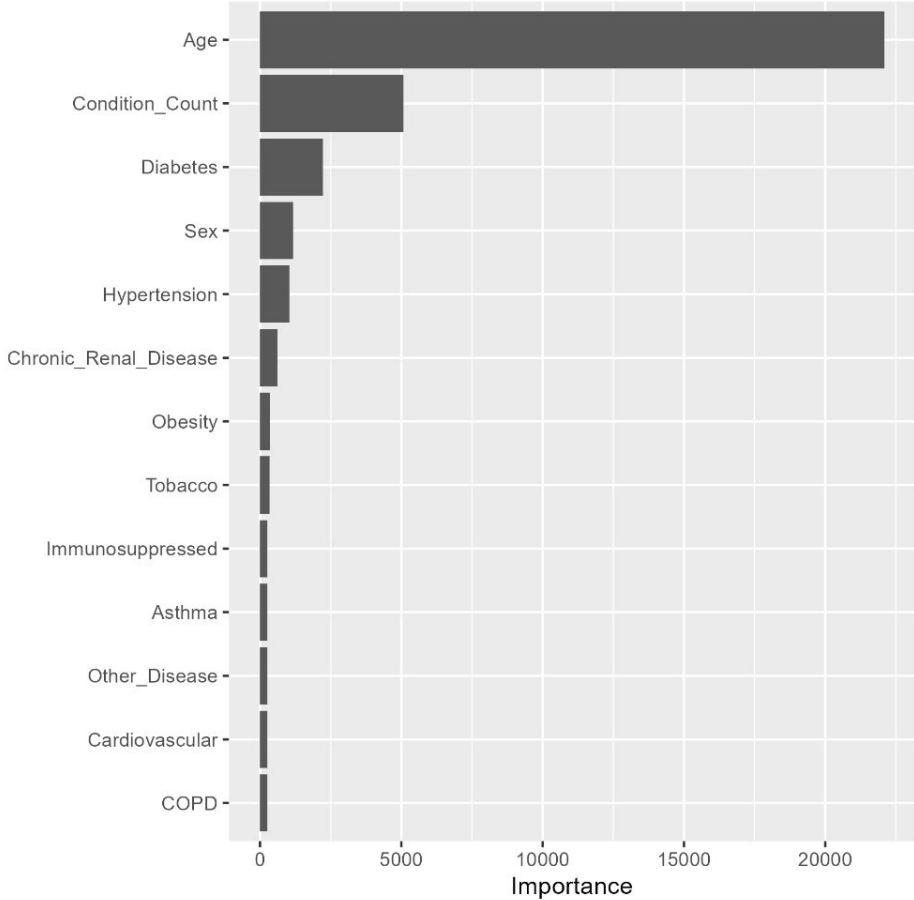


7.3.2 LR - Reduced Dataset

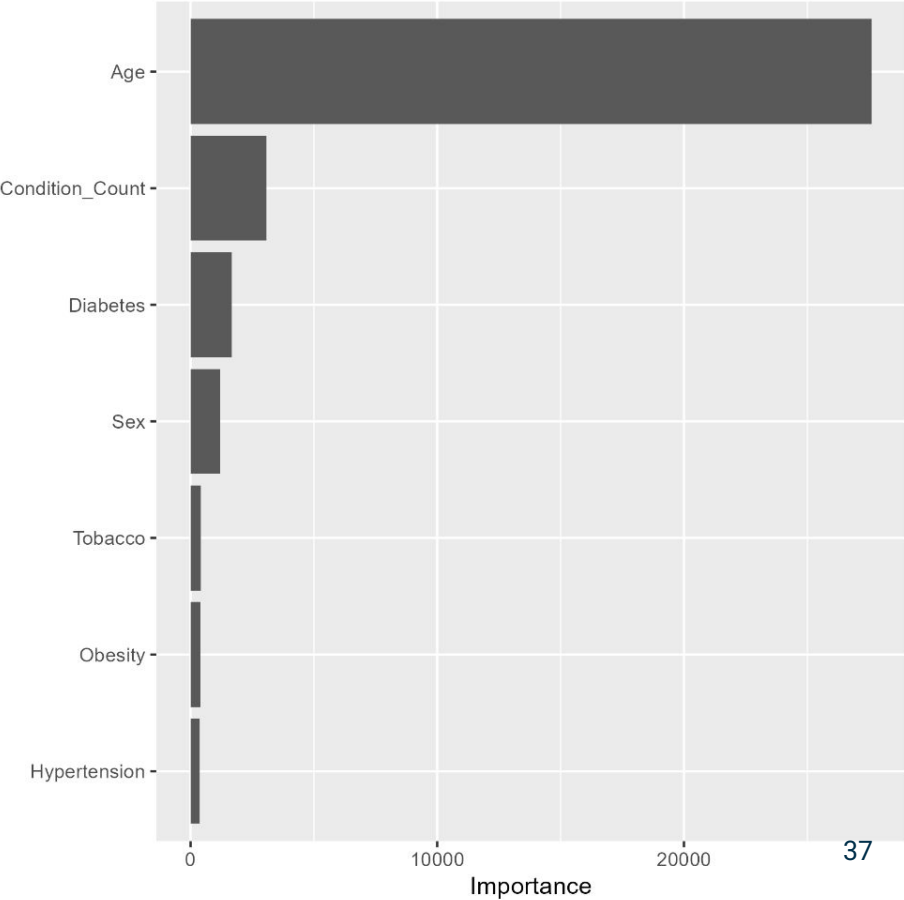


Random Forest

7.3.3 RF - Full Dataset



7.3.4 RF - Reduced Dataset



Key Takeaway

Older age, higher condition burden, male sex, diabetes, hypertension, and chronic renal disease were the most important predictors.



Logistic Regression Results

- **4+ conditions:** 5.84x higher odds of adverse outcome
- **3 conditions:** 4.32x higher odds
- **1 condition:** ~2.0x higher odds
- **Chronic renal disease:** ~2.0x higher odds
- **Male sex:** 1.87x higher odds



Important Caution

Tobacco and asthma showed lower estimated odds in this model, but should **not** be interpreted as protective.

Variable Importance Across Models:

Age ranked highest, followed by condition count, diabetes, sex, and hypertension.

Conclusion

Limitations, Bias & Model Constraints

Conclusion – Limitations & Bias

Sampling bias

Confirmed Cases:

Only confirmed cases from medical units → missing untested individuals or tested outside monitoring system

Outcome Bias:

Likely overestimates severe outcomes

Class imbalance

~15% Proportion

Metric Shift:

Accuracy is less meaningful

Key Focus:

Prioritize sensitivity / F1 scores

Limited variables

Clinical Gaps:

Missing clinical data (labs, symptoms)

Exclusions:

Conditions like mental health or cancer not included

Conclusion – Model Constraints

Measurement Error

Conditions likely **self-reported**, leading to potential underreporting of pre-existing health issues

Outdated Data:

From first year of the pandemic

Outcome Misclassification

Patients initially "sent home" may later experience worsening symptoms, impacting label accuracy

Model Limitations

Logistic Regression:

Limited ability to capture complex nonlinear patterns

Random Forest:

Highly predictive but significantly less interpretable

Conclusion – Research Questions

1. Risk Factors by Age

Strongest Predictor:

Age group, followed by total condition burden

Key Conditions:

Diabetes, hypertension, and sex

Note: Males show higher risk than females

2. Model Comparison

Logistic Regression:

High accuracy (~85%), but low sensitivity and poor at identifying adverse outcomes

Random Forest:

Better sensitivity and F1 score (~0.49 vs. ~0.31) despite slightly lower accuracy (~81%)

3. Comorbidity Interactions

Diabetes & Hypertension show consistent strong associations

Key Conditions:

Chronic Renal Disease had highest lifts across age groups

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Sertbas, Y. E. (2023). "Clinical outcomes of COVID-19 in patients with". *North Clinical Istanbul*, 401-410. Retrieved from "Clinical Outcomes of COVID-19 in Patients with Chronic Diseases.": https://pdf.journalagent.com/nci/pdfs/NCI-64436-RESEARCH_ARTICLE-SERTBAS.pdf

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Thank You

Questions?

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